

Mixtures & Alligations

1. In what ratio must wheat costing ₹14.4 per kg be mixed with wheat costing ₹20.4 per kg so that the mixture is worth ₹18 per kg?
(1) 1:1 (2) 2:3 (3) 3:2
(4) 3:4 (5) 1:2
2. A solution of 60 litres of milk and water contains 20% water. How much water should be added to this solution, to make it a solution with 40% quantity of water?
(1) 10 litres (2) 15 litres (3) 20 litres
(4) 25 litres (5) 30 litres
3. A mixture of 60 litres of milk and water contains 20% water. How much pure milk should be added to the mixture to make it a mixture with 15% water?
(1) 18.75 litres (2) 24 litres
(3) 17.5 litres (4) 20 litres
4. Milk and water are mixed in the ratio 5: 4 in a can. If by adding 5 litres of water the can becomes full and the ratio of milk to water changes to 4: 5, what is the capacity of the can?
(1) 20 litres (2) 27.5 litres
(3) 25 litres (4) 22.5 litres
5. Two vessels contain mixtures of milk and water. The ratio of milk and water is 4: 3 in the first vessel and 5: 6 in the second vessel. In what ratio should these two mixtures be mixed such that the resultant mixture has milk and water in the ratio 6:5?
(1) 7:2 (2) 5:2 (3) 4:1
(4) 6:1 (5) 3:1
6. How many litres of a mixture having spirit and water in the proportion of 3:2 should be added to 60 litres of 75% spirit solution such that a resultant mixture having $\frac{2}{3}$ rd of spirit is obtained?
(1) 50 litres (2) 60 litres
(3) 75 litres (4) 80 litres
7. A can contains 50 litres of pure milk. From this, 5 litres of milk is removed and 5 litres of water is added. Again 5 litres of the solution is removed and 5 litres of water added. What is the quantity of milk in the mixture now?
(1) 40.5 litres (2) 39.5 litres
(3) 41.5 litres (4) 42.5 litres
8. From a 500 ml of pure milk solution, 10% of milk is removed and replaced with water. Again, 10% of the solution is replaced with water. What is the ratio of milk and water in the final solution?
(1) 9:1 (2) 7:3 (3) 81:19
(4) 79:21 (5) 91:9
9. How many litres of a dye solution with 78% concentration be mixed with 24 litres of a dye solution with 64% concentration to get a resultant solution of with 70% concentration?
(1) 18 litres (2) 32 litres (3) 24 litres
(4) 20 litres (5) 15 litres
10. Two vessels contain spirit in 0.5 and 0.75 concentration. If 2 litres from the first and 3 litres from the second are mixed, what will be the ratio of spirit and water in the resultant solution?
(1) 13:7 (2) 7:17
(3) 15:17 (4) 17:15
11. A dealer mixes tea costing ₹18 a kg with a high quality tea and sells the mixture at ₹24 a kg. Thus earning a profit of 20%. If the ratio in which the two quality tea were mixed is 1: 1, what is the cost of the higher quality tea?
(1) ₹25 per kg. (2) ₹26 per kg.
(3) ₹24 per kg (4) ₹22 per kg.

12. The concentration of spirit is 25%, 35% and 55% in three different vessels. If 3 litres from the first, 6 litres from the second and 1 litre from the third are mixed, what would be the resultant concentration of spirit?

(1) 27% (2) 34%
(3) 36% (4) 45%

13. Three vessels contain mixtures of milk and water, the ratio of capacities of the three vessels are in the ratio 5:6: 8. The ratio of milk to water in each of the three vessels is respectively 3:2, 4:1 and 2: 1. What is the ratio of milk to water in the resultant mixture obtained by adding a fifth of the first vessel, one half of the second and a fourth of the third vessel?

(1) 13:5 (2) 65:37
(3) 37:40 (4) 32:67

14. In three milk solutions, the quantities of milk and water are in the ratio 5:3, 5: 1 and 7: 5. If the solutions are mixed in the ratio 4: 3 : 2, then what will be the ratio of milk and water in the final mixture?

(1) 26:19 (2) 19:17 (3) 26:17
(4) 17:19 (5) 37:17

15. A sum of ₹693 was collected for a charity, from boys and girls of a class of 45 students. If each boy contributes ₹15 and each girl contributes ₹16, find the number of girls in the class.

(1) 24 (2) 22 (3) 20
(4) 18 (5) 16

Analytical Puzzles - Seating Arrangement - Linear**Directions for questions 1 to 3:**

Five boys A to E sit in a row facing north. They sit in five different positions (1 to 5) from left to right, not necessarily in the same order.

- i) Two boys sit in between A and D.
- ii) C sits exactly in the middle of the row but not adjacent to A.
- iii) B sits to the right of E.

1. What (is/are) the order/s in which these five boys sit (from left to right)?

- a) EDCAB b) EDCBA c) AECDB
- d) ABCDE e) Both (b) and (c)

2. How many boys sit between A and E?

- a) One b) Two c) Three
- d) (a) or (b) e) Cannot be determined

3. Who is to the immediate left of D?

- a) B b) C c) A
- d) E e) Cannot be determined

Directions for questions 4 and 5:

Five persons (A to E) sit in a row facing south, not necessarily in the same order.

- i) B is to the right of D and is not at any one of the ends.
- ii) C is to the left of A.
- iii) D is to the immediate left of E who is at the centre.

4. Who is to the immediate left of B?

- a) C b) D c) A
- d) E e) Cannot be determined

5. What is/are the orders in which these five persons can sit (from right to left)?

- a) ACEDB b) CAEBD c) ABECD
- d) ABEDC e) Cannot be determined

Directions for questions 6 and 7:

Six persons – A to F stand in a queue, but not necessarily in the same order.

- i) Exactly two persons stand ahead of D and exactly two behind C.
- ii) Two persons stand between A and E.
- iii) A stands exactly between C and F.

6. Who stands ahead of D but behind B?

- a) C b) A c) E
- d) C or A e) A or F

7. How many persons stand behind B?

- a) One b) Two c) Three
- d) Four e) Five

Directions for questions 8 to 12:

Seven persons Esha, Fathima, Ganesh, Hari, Indu, Jothi and Kavin sit in a row facing North, but not necessarily in the same order. The following information is known about them.

There are three persons who sit between Esha and Jothi. The number of persons who sit to the left of Esha is equal to the number of persons who sit to the right of Jothi. Either Fathima or Indu sits adjacent to Jothi. Kavin sits second to the right of Hari. Ganesh sits at an end and sits second to the right of Fathima.

8. Who sits in the middle of the row?

- a) Kavin b) Indu c) Fathima
- d) Jothi e) Esha

9. What is the position of Ganesh with respect to Jothi?

- a) Second to the left b) Immediate left
- c) Third to the left d) Immediate right
- e) Second to the right

10. Who sits at the right end?

- a) Hari b) Ganesh c) Jothi
- d) Esha e) Kavin

11. How many persons sit between Indu and Ganesh?

- a) 1 b) 2 c) 3 d) 4 e) 5

12. Which of the following is/are definitely false?

- a) Esha sits to the immediate left of Hari.
b) Ganesh and Hari are at the ends.
c) Kavin and Fathima are adjacent to Jothi.
d) None of these
e) More than one of the above

Directions for questions 13 to 17:

Nine persons A through I, sit in a row. Some are facing north and the rest are facing South.

The person who sits exactly in the middle of the row is facing South. I is three places away from A who sits fourth to the right of G. D sits at an end and is to the immediate right of I, who faces North.

E is adjacent to both A and B. B is third from an end, and is second to the right of G. C is adjacent to neither B nor A. C and H are facing the same direction and are opposite to the direction E and D face. E and A face opposite direction. F is second to the left of E. F is to the immediate right of B and is facing north.

13. Who sits in the middle of the row?

- a) G b) I c) A d) E e) H

14. Who sits to the immediate left of A?

- a) E b) B c) H d) I e) None of these

15. How many persons are facing North?

- a) Five b) Four c) Three
d) Six e) Cannot be determined

16. Four of the following five pairs are alike in a certain way based on the given arrangement. Find the one that does not belong to the group.

- a) G, B b) E, H c) H, I
d) B, A e) C, D
d) F is two places away from G.
e) More than one of the above.

17. Which of the following statements is/are true?

- a) G and F face the same direction.
b) C and I face different directions.
c) H sits exactly between E and C.

Directions for questions 18 to 22:

Eight persons- I, J, K, L, M, N, P, and Q – are standing in a queue, not necessarily in the same order. The following information is known about them. Only two persons are standing between I and L. I stands just behind P. There are three persons standing between L and M. I is not standing adjacent to M. K stands adjacent to J and N. At least one person stands between L and J.

18. Who stands just behind L?

- a) K b) N c) P
d) Q e) Cannot determined

19. How many persons stand between Q and P?

- a) One b) Zero c) Two
d) Three e) None of these

20. Who stand(s) exactly between K and P?

- a) Q b) J c) N d) L e) Q and I

21. Four of the following are alike in a certain way and form a group. Find the one which does not belong to that group.

- a) M, J b) N, L c) Q, P
d) P, M e) K, N

22. How many persons stand in front of K, if L and K interchange their positions?

- a) One b) Two c) Three
d) Four e) More than four